

KIRUNA, Sweden

WMO: 020440

Lat: 67.823N

Lon: 20.340E

Elev: 1466

StdP: 13.93

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.2	-14.7	-26.8	1.3	-20.3	-20.6	1.9	-14.5	29.2	26.9	25.6	24.2	3.2	220	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	73.6	58.4	69.9	57.1	66.2	55.7	61.2	69.9	59.1	66.2	57.2	64.1	7.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.9	75.7	63.6	55.8	70.0	62.0	53.8	65.0	60.1	27.9	70.4	26.5	66.3	25.2	64.2	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.4	20.1	17.3	DB	-25.9	77.7	5.8	3.3	-30.0	80.1	-33.4	82.0	-36.6	83.9	-40.8	86.3	(4)
			WB	-26.2	63.6	5.7	2.1	-30.3	65.1	-33.6	66.3	-36.8	67.5	-40.9	69.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	31.4	9.9	11.2	18.2	29.0	40.0	49.6	55.9	51.8	43.6	30.9	20.3	15.0	(6)
(7)		DBStd	18.28	13.04	13.46	9.95	7.55	7.44	6.82	5.90	5.66	6.14	8.43	9.70	11.65	(7)
(8)		HDD50	7185	1242	1085	984	629	324	89	13	45	206	593	891	1084	(8)
(9)		HDD65	12268	1707	1505	1449	1079	775	463	288	411	643	1058	1341	1549	(9)
(10)		CDD50	400	0	0	0	0	14	77	196	100	13	0	0	0	(10)
(11)		CDD65	8	0	0	0	0	0	1	6	1	0	0	0	0	(11)
(12)		CDH74	85	0	0	0	0	3	14	60	8	0	0	0	0	(12)
(13)		CDH80	4	0	0	0	0	0	1	3	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.3	6.7	7.5	8.6	7.8	7.7	6.7	6.3	7.4	7.2	6.8	7.0	(14)	
(15)	Precipitation	PrecAvg	19.8	1.1	0.9	0.9	1.1	1.3	2.1	3.3	2.9	1.8	1.6	1.5	1.3	(15)
(16)		PrecMax	25.8	2.8	1.9	2.2	2.6	3.7	7.0	5.9	5.9	3.3	3.5	3.6	3.3	(16)
(17)		PrecMin	13.6	0.4	0.2	0.1	0.1	0.1	0.3	0.9	0.6	0.6	0.3	0.6	0.5	(17)
(18)		PrecStd	3.6	0.6	0.5	0.6	0.7	0.8	1.4	1.4	1.4	0.8	0.8	0.7	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	41.3	41.8	52.6	72.0	75.5	80.0	74.6	64.2	51.1	43.2	41.7	(19)
MCWB			36.4	36.1	35.8	43.6	53.1	57.9	62.5	60.6	53.7	46.1	39.3	37.4	(20)	
2%		DB	35.5	37.1	37.8	46.3	64.0	70.2	75.3	70.3	59.2	48.2	38.6	36.9	(21)	
		MCWB	32.2	33.3	33.5	38.9	50.2	54.8	59.3	58.9	51.7	44.6	36.2	33.5	(22)	
5%		DB	30.7	33.2	35.4	43.1	57.4	66.1	71.4	66.1	56.1	45.8	35.4	33.1	(23)	
		MCWB	28.7	30.2	31.8	37.2	47.1	53.3	58.7	57.2	50.3	42.7	33.5	30.9	(24)	
10%		DB	27.1	29.5	32.3	40.4	52.2	62.3	67.4	62.4	53.6	42.5	33.0	30.2	(25)	
		MCWB	25.4	27.6	29.3	35.5	44.0	51.9	57.1	54.9	48.9	40.0	31.6	28.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.7	37.0	36.3	43.6	54.6	61.2	64.7	63.6	55.8	47.5	39.5	37.7	(27)
MCD B			40.7	41.3	40.9	52.1	68.2	71.3	73.5	72.0	60.7	49.2	42.5	41.3	(28)	
2%		WB	32.4	33.2	34.0	39.8	51.0	57.6	62.0	60.2	53.3	44.9	36.2	33.9	(29)	
		MCD B	35.0	36.8	37.4	44.7	62.2	65.3	71.3	67.5	57.4	47.8	38.2	36.1	(30)	
5%		WB	28.8	30.3	31.9	37.7	47.5	55.0	60.1	58.0	51.4	42.5	33.6	30.9	(31)	
		MCD B	30.8	32.6	34.9	42.4	56.7	63.6	68.2	64.5	55.2	45.5	34.9	32.7	(32)	
10%		WB	25.5	27.4	29.5	35.7	44.8	52.6	58.3	55.8	49.6	39.9	31.6	28.5	(33)	
		MCD B	27.1	29.2	32.3	40.1	51.3	60.8	65.3	61.3	53.0	42.0	32.7	30.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.3	14.4	16.1	14.8	13.8	14.4	15.1	14.8	13.1	10.3	11.5	13.5	(35)
MCD BR			14.2	15.4	14.7	15.9	22.7	20.8	21.2	21.3	16.9	12.5	11.5	13.4	(36)	
MCWBR			12.7	12.9	12.1	10.7	12.4	9.8	9.5	11.5	10.8	9.4	10.0	11.7	(37)	
5% WB		MCD BR	14.1	13.8	14.0	14.7	20.5	18.1	17.7	19.0	14.3	11.7	10.5	12.8	(38)	
		MCWBR	12.6	11.7	11.8	10.1	11.8	9.7	9.0	11.1	9.8	9.1	9.4	11.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.169	0.230	0.269	0.297	0.317	0.322	0.328	0.315	0.284	0.252	N/A	N/A	(40)	
(41)		taud	1.933	2.141	2.232	2.268	2.451	2.520	2.516	2.532	2.585	2.351	N/A	N/A	(41)	
(42)		Ebn at Noon	47	190	248	272	279	281	274	266	250	184	N/A	0	(42)	
(43)		Edh at Noon	3	14	24	31	29	28	27	24	17	12	N/A	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	13	158	624	1230	1570	1584	1478	1068	584	195	32	0	(44)	
(45)		RadStd	1	14	30	56	134	123	143	89	69	28	3	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page